

Bio-Fuels

Qualification Pack: SGJ/Q4102 | **NSQF Level:** 4 | **Total Duration:** 30 Hours (1,800 Mins)

Module 1: Entrepreneurship and Basics of Biomass Energy (3.0 Hours)

UNIT 1.1 Fundamentals of Biofuels & Biomass Energy (0.75 Hours / 45 Mins)

- 1.1.1 Biofuels definition, role in sustainable energy & 1st to 4th generation fuels
- 1.1.2 Types of biofuels: Solid (firewood, pellets), Liquid (ethanol, biodiesel), Gaseous (biogas, syngas)
- 1.1.3 Biomass in global energy transition & carbon-neutral concept
- 1.1.4 Conversion technologies: Combustion, digestion, gasification, pyrolysis

UNIT 1.2 Entrepreneurship in Biomass Sector (0.75 Hours / 45 Mins)

- 1.2.1 Concept of entrepreneurship development & rural economic impact
- 1.2.2 7 steps of the entrepreneurial process (Opportunity identification to investment harvesting)
- 1.2.3 Business Models (Biomass-to-Energy, Biofuels) vs. Business Strategies (Cost leadership, differentiation)

UNIT 1.3 Managing a Biomass Enterprise (0.50 Hours / 30 Mins)

- 1.3.1 Management challenges in emerging MSMEs & startups
- 1.3.2 Addressing Financial, HR, Product/Service Development, and Operations risks

UNIT 1.4 Biomass Resource Management & Procurement (1.00 Hour / 60 Mins)

- 1.4.1 Biomass procurement systems & supply chain challenges (seasonality, bulk transport)
- 1.4.2 Agro-residue waste management (Primary vs. Secondary residues)
- 1.4.3 Field survey methodology with farmers, aggregators, and FPOs
- 1.4.4 Resource potential mapping, pricing, logistics, cropping & harvesting patterns

Module 2: Manage Financial Budget & Develop Business Plans (6.0 Hours)

UNIT 2.1 Fundamentals of Financial Management (1.50 Hours / 90 Mins)

- 2.1.1 Accrual vs. Cash accounting, CAPEX vs. OPEX, asset valuation
- 2.1.2 Structure & interpretation of Balance Sheet, Income Statement, Cash Flow Statement
- 2.1.3 Break-Even Point (BEP) analysis, formulas, and practical applications

UNIT 2.2 Financing Strategies for Biomass Ventures (1.50 Hours / 90 Mins)

- 2.2.1 Stages of financing (Pre-seed, Seed, Startup, Growth, Maturity, Renewal)
- 2.2.2 Debt vs. Equity financing characteristics, risk profiles, and control
- 2.2.3 Funding sources: Bank loans, Green Bonds, VCs, Angel Investors, Grants, PPPs
- 2.2.4 Managing firm cash flows & seasonal procurement costs

UNIT 2.3 Business Development & Innovation (1.00 Hour / 60 Mins)

- 2.3.1 8 Core business development tasks & market research
- 2.3.2 Sustainable Growth Rate (SGR) model derivation & financing gap calculations
- 2.3.3 10 Characteristics of a good innovation for business plan development

UNIT 2.4 Operations & Technology in Plants (1.00 Hour / 60 Mins)

- 2.4.1 Pellet plant operational workflow: Receiving, drying, grinding, pelletizing, cooling, storage
- 2.4.2 Identification & operation of machinery (hammer mill, pellet mill, digester, gas storage)
- 2.4.3 Technology selection criteria based on feedstock, scale, and efficiency

UNIT 2.5 Supply Chain, Storage, & Budgeting (1.00 Hour / 60 Mins)

- 2.5.1 Demand & supply analysis, catchment area selection
- 2.5.2 Pellet storage options & shelf-life factor management (moisture < 10%)
- 2.5.3 Biogas storage options (low, medium, high pressure)
- 2.5.4 Budgetary cost estimation for pellet plants and small biogas units

Module 3: Manage Sales, Supply, and Marketing of Product (3.0 Hours)

UNIT 3.1 Marketing Strategies for Pellets & Biogas (0.75 Hours / 45 Mins)

3.1.1 Sales promotion tools (Consumer, Trade, and Digital promotions)

3.1.2 4Ps of Product Marketing vs. 7Ps of Service Marketing in bioenergy

UNIT 3.2 Sales Planning & Customer Engagement (0.75 Hours / 45 Mins)

3.2.1 E-commerce scope, digital sales channels, and online marketplaces

3.2.2 Biogas & pellet sales estimation methods (consumption-based & market trends)

3.2.3 Step-by-step Sales Process: Lead generation to closing & post-sales support

3.2.4 Customer satisfaction & service tracking techniques (NPS, CRM, surveys)

UNIT 3.3 Market Research & Performance Assessment (0.75 Hours / 45 Mins)

3.3.1 Global marketing potential for renewable energy MSMEs

3.3.2 Criteria to assess marketing performance (sales growth, market share, ROI)

3.3.3 Customer profiling via Primary & Secondary research (power plants, industries, households)

UNIT 3.4 Organic Farming Economics & Sales Models (0.75 Hours / 45 Mins)

3.4.1 Cost of production: Conventional vs. Organic farming systems

3.4.2 Profitability estimation for direct-to-consumer organic by-product sales

3.4.3 Procedures for selling via local markets, organic huts, and farmgate channels

Module 4: Compliance with Applicable Laws, Policies & Procedures (3.0 Hours)

UNIT 4.1 Regulatory Framework & Authorities (0.75 Hours / 45 Mins)

4.1.1 Key regulatory authorities (SPCB, SERC, SNA, ULBs, Panchayats)

4.1.2 Statutory laws (Environment Protection Act, Air/Water Acts) & BIS/ISO standards

UNIT 4.2 Environmental Regulations & Pollution Control (0.75 Hours / 45 Mins)

4.2.1 Waste Management Rules (Solid Waste, Hazardous Waste Rules 2016)

4.2.2 Procedures to obtain SPCB clearances: Consent to Establish (CTE) & Consent to Operate (CTO)

UNIT 4.3 Business Registration & Legal Formalities (0.75 Hours / 45 Mins)

4.3.1 Government coordination & business registration methods (SPICe+, GST, MSME/Udyam)

4.3.2 Record-keeping best practices, digital filing, and audit readiness

UNIT 4.4 Government Schemes for Bio-Energy (0.75 Hours / 45 Mins)

4.4.1 National Bio-Energy Programme, SATAT, GOBARdhan, and Biomass Power schemes

4.4.2 Eligibility criteria, subsidies, financial benefits, and application workflows

Module 5: Manufacturing of Biomass Pellets (Elective 1) (3.0 Hours)

UNIT 5.1 Biomass Preparation & Die Placement (0.75 Hours / 45 Mins)

5.1.1 Biomass quality preparation (crushing < 5mm, moisture 10-15%)

5.1.2 Methods of biomass placement on dies & extrusion mechanics under heat/pressure

UNIT 5.2 Machinery Operations in Pellet Production (0.75 Hours / 45 Mins)

5.2.1 Operations of dryers, hammer mills, pellet mills, coolers, and bomb calorimeters

5.2.2 Speed, temperature, and moisture control parameters

5.2.3 Cooling mechanics, backpressure management, and 3mm die adjustments

UNIT 5.3 Quality Control & Testing of Pellets (0.75 Hours / 45 Mins)

5.3.1 Standard testing protocols: GCV (Bomb Calorimeter), Ash content, Moisture content

5.3.2 Operating moisture meters for agricultural waste quality verification

UNIT 5.4 Post-Production Maintenance & Best Practices (0.75 Hours / 45 Mins)

5.4.1 Post-production cleaning, LOTO protocols, and machinery maintenance schedules

5.4.2 Essential Do's and Don'ts in biomass pellet manufacturing

Module 6: Installation & Operation of Small Biogas Plant (Elective 2) (6.0 Hours)

UNIT 6.1 Introduction & Comparative Analysis (1.00 Hour / 60 Mins)

6.1.1 Overview of anaerobic digestion & biogas generation mechanics

6.1.2 Comparative analysis: Biogas vs. LPG vs. CNG (sources, composition, uses)

6.1.3 Environmental & economic benefits of biogas technology

UNIT 6.2 Site Assessment & Material Selection (1.50 Hours / 90 Mins)

6.2.1 Site assessment factors (terrain, distance, water access, sunlight, safety)

6.2.2 Sizing biogas plants based on feedstock, household size, and cattle availability

6.2.3 Selecting materials (bricks, cement, pre-fabricated modular components)

UNIT 6.3 Construction & Installation of Biogas Plant (1.50 Hours / 90 Mins)

6.3.1 Site preparation: Layout marking, excavation, leveling, bed concrete casting

6.3.2 Civil construction of digester tank, inlet/outlet chambers, and gas holder

6.3.3 Gas distribution pipeline layout, regulator, and stove installation

UNIT 6.4 Operation, Maintenance, & Safety (1.00 Hour / 60 Mins)

6.4.1 Gas leakage testing protocols (soap solution & pressure detectors)

6.4.2 Plant charging (1:1 cattle dung to water mix) and regular feeding management

6.4.3 Routine maintenance of pipelines, moisture traps, and gas apparatus

6.4.4 Dos and Don'ts of safe gas usage

UNIT 6.5 Utilization of Biogas & Bio-Slurry (1.00 Hour / 60 Mins)

6.5.1 Methane as a green substitute for LPG, wood, cattle dung, and kerosene

6.5.2 Bio-slurry drying (sun vs. mechanical), moisture control, and packaging as compost

6.5.3 Application techniques of bio-slurry in agricultural fields and fishery ponds

Module 7: HSE (Health, Safety & Environment) (3.0 Hours)

UNIT 7.1 Workplace Safety & Hazard Identification (0.75 Hours / 45 Mins)

7.1.1 Safe work area requirements (exits, ventilation, fire safety zones)

7.1.2 Occupational health & safety standards (OSHA, Factories Act)

7.1.3 Work safety procedures for operating bioenergy machinery

UNIT 7.2 First Aid & Emergency Response (0.75 Hours / 45 Mins)

7.2.1 Administering first aid for cuts, thermal/chemical burns, electric shocks

7.2.2 Emergency protocols for fires, gas leaks, and plant evacuations

UNIT 7.3 Safety Practices & Equipment Protection (0.75 Hours / 45 Mins)

7.3.1 PPE selection and correct usage (gloves, goggles, respirators, safety shoes)

7.3.2 Electrical & mechanical equipment protection, and Lockout/Tagout (LOTO) procedures

UNIT 7.4 Waste Management & Housekeeping (0.75 Hours / 45 Mins)

7.4.1 5S Methodology for facility housekeeping

7.4.2 Infection control guidelines & biohazard handling

7.4.3 Safe disposal of ash, chemical residue, and industrial waste

Module 8: Employability Skills (3.0 Hours)

UNIT 8.1 Professional & Workplace Skills (1.00 Hour / 60 Mins)

8.1.1 Constitutional values, citizenship, environment sustainability, 21st-century skills

8.1.2 Basic English communication skills for everyday workplace interactions

UNIT 8.2 Financial, Digital, & Legal Literacy (1.00 Hour / 60 Mins)

8.2.1 Essential digital skills, internet safety, word processing & spreadsheet basics

8.2.2 Financial literacy, salary components, taxes, and transaction security

8.2.3 Legal rights, POSH Act awareness, gender sensitivity, and inclusion

UNIT 8.3 Career Planning & Job Readiness (1.00 Hour / 60 Mins)

8.3.1 Career planning, goal setting, professional resume creation, and interview readiness

8.3.2 Customer service etiquette and grievance redressal

8.3.3 Apprenticeship opportunities & portal registration procedures